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SHAPLEY ENSEMBLE ADVERSARIAL ATTACK

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ABSTRACT

An intuitive strategy for generating more transferable adversarial examples is to absorb the advantages of different surrogates in an ensemble. However, existing ensemble adversarial attacks simply average the outputs of various models while ignoring their different contributions. To quantify the importance of various surrogates, we propose a novel Shapley Ensemble Adversarial Attack (dubbed **SEAA**) – an effective algorithm that allocates weights based on Shapley values of the surrogates. Specifically, the synthesis of adversarial examples in an ensemble adversarial attack is firstly regarded as a cooperative game process of multiple surrogates from the perspective of contribution. As the Shapley value is known as an effective decision metric in cooperative game theory, it is thus intuitively introduced into this work to accurately evaluate the contribution of a surrogate by reweighing its importance at each iteration, thus avoiding local optimality and steering the generation of adversarial examples. Comprehensive experimental results demonstrate our effectiveness.

***Index Terms*—** ensemble adversarial attack, transferability, Shapley value, black-box attack

1. INTRODUCTION

Adversarial examples attract enormous attention since they can be applied to reveal defects of DNNs for promoting their robustness. Till now, most of existing efforts for the generation of adversarial examples are based on optimization or gradient, such as Fast Gradient Sign Method (FGSM) [1] and its various variants [2]. They easily achieve high success rates of attack under the impractical white-box setting, while suffering from unsatisfactory performance in more realistic black-box attacks caused by poor transferability.

Many of the black-box methods are transfer-based attacks [2, 3]. Their core idea is to attack the proxy model at hand under the white-box setting for the generation of adversarial examples, and then the results can be transferred to the target model. However, a single proxy model significantly limits the transferability of these adversarial examples, albeit the improvement of attack performance. That is, the success rate of attack will be affected, if the proxy model is greatly different from the target model. To this end, an intuitive solution is ensemble adversarial attacks, where several white-box surrogate models are analogized to the training data and then the generation of adversarial examples is treated as the parameter update of DNNs [4, 2]. Hence, the transferability on black-box models can be thought of as the generalization of neural networks into the test data [4]. Early ensemble approaches [2, 5] simply average the outputs of several surrogate models over losses, logits [2] or just predictions [5]. Such a plain fusion addresses the diversity of models yet ignores the different properties between them. Xiong *et al.* recently introduce the SVRE optimizer [4] to reduce gradients variance of various models during optimization, and Adaptive model ensemble adversarial attack (AdaEA) [6] proposes a discrepancy ratio and a disparity-reduced filter to reduce the diversity between surrogates. Nevertheless, a recent study [7] argues that the variance of gradient is not always correlated with the generalization and the existing methods neglect different contributions of various surrogates.

To remedy such shortcomings, we argue that different surrogate models should be assigned different weights instead of treated uniformly during optimization. To quantify the individual importance of various surrogates, we propose a novel Shapley Ensemble Adversarial Attack (dubbed **SEAA**) method, which is an effective algorithm to allocate the weight based on their Shapley values over the prediction. Specifically, the synthesis of adversarial examples in an ensemble attack is regarded as a cooperative game process of multiple surrogates from the perspective of contribution. As the Shapley value is known as an effective decision metric in cooperative game theory, it is thus introduced into this work to accurately evaluate the contribution of a surrogate. Based on that, we can intuitively reweigh the importance of a surrogate at each iteration, thus avoiding local optimal and steering the generation of adversarial examples.

Our major contributions can be concluded as: 1) We im-

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prove the transferability of ensemble adversarial attacks from the perspective of cooperative game theory by treating each surrogate as a player. 2) We introduce the Shapley value into an ensemble to accurately quantify the contribution of each surrogate model. 3) Comprehensive experiments under different settings confirm our superiority and effectiveness.

2. RELATED WORK

Adversarial Attack based on transfer can be divided from two aspects: gradient-based optimization [1, 2] and input transformation [3, 8]. The former [2] attempts to boost the transferability by formulating various advanced gradient optimization strategies based on Iterative Fast Gradient Sign Method (I-FGSM) [9], while the second category enhances the transferability by transforming the input images, which is similar to the data augmentations, such as Diverse Input Method (DIM) [8] and Translation-Invariant (TI) attack [3].

However, they all suffer from a common disadvantage that the generated adversarial examples heavily depend on some model, and thus such homogeneity may limit the improvement of performance. An intuitive solution is ensemble adversarial attacks, where several white-box surrogates are analogized to the training data and then the generation of adversarial examples is treated as the parameter update of DNNs [2].

Ensemble Adversarial Attack is an effective strategy to boost the transferability for two reasons: (1) This can reduce the generalization error upper bound of the empirical risk minimization [10]; (2) It can also be interpreted as the enlarging of training samples in deep learning. The following efforts [2, 5] still simply fused the logits of surrogates or just averaged over losses to formulate an ensemble. Other efforts [10] attempted to leverage existing surrogates to create more variants, but they also just averaged their logits. As aforementioned, such a superficial fusion overlooks different properties of surrogates. The latest works [4, 6] tune the gradient variance or reduce their diversity over surrogate models.

However, indiscriminately treating all surrogates will confuse the optimization, or only focusing on variance reduction still limits the transferability of the generated adversarial examples. Differently, this work first views an ensemble adversarial attack as a cooperative game process, where all surrogate models are treated as players with different contributions. Therefore, maximizing the collective contribution is positively correlated with the performance of adversarial examples. Then, Shapley value is introduced to accurately evaluate the contribution of a surrogate, and we can reweigh its importance at each iteration to steer the generation toward enhancement of adversarial transferability.

Shapley Value has been well studied in the cooperative game theory as a fair contribution distribution method [11]. Several works [11] aim to explain the feature importance in DNNs or multi-agent reinforcement learning. Some other works [12] attempt to model the importance of different components. Note that Shapley value computation is NP-hard, some efficient strategies [12, 11, 13] are thus successively introduced for fast approximation.

We extend the Shapley value to ensemble adversarial attack, so that the optimal weights are allocated to different surrogate models during each iteration, which thus contributes significantly to the transferability of adversarial examples.

3. METHODOLOGY

We treat each surrogate in an ensemble adversarial attack as a player and introduce the cooperative game to boost the adversarial transferability. Subsequently, the Shapley value is applied to accurately estimate the individual contribution to the cooperation. Thus the importance of each surrogate can be accordingly reweighed at each iteration rather than treated equally across all surrogates throughout the optimization like existing SVRE [4] and plain average [2]. For brevity, our work mainly focuses on the naive Shapley estimation to demonstrate our superiority.

Shapley Value. Due to the existence of complicated interactions among surrogate models, it is difficult to metric the individual contribution of each surrogate to the final generation of adversarial examples. To solve this issue, we intuitively view the ensemble adversarial attack as a multi-player game, where each surrogate model is defined as a player. In cooperative game theory, the Shapley value is one of the most important tools to determine the contribution of each player to the overall output by allocating values to each player with the average of marginal values. Let m be a metric function to evaluate the performance of players, which can be attack success rate or loss increment in the adversarial attack. Thus, the marginal contribution of i-th surrogate model can be defined as: $margin(i) = m(S \cup i) - m(S)$, where S is a subset of surrogate models excluding i-th model, indicated as $S \subseteq F \setminus i$. F denotes the set of all the surrogate models, and $N = |F|$ means the number of surrogates in F. Then we can formulate the Shapley value ϕ for model i as the average of the marginal contributions, defined as:

$$\phi_i(m) = \frac{1}{N} \sum_{S \subseteq F \setminus i} \pi_S \cdot (m(S \cup i) - m(S)), \tag{1}$$

where $\pi_S = \frac{(N-|S|-1)!|S|!}{(N-1)!}$ indicates the relative importance of subset S, and $|S|$ means the cardinality of S.

Estimation for Shapley Value. It can be seen from Eq.(1) that the Shapley value is the average of marginal contributions of the surrogate model in all possible orders. For the black-box attack, we can not access anything about the victim model. Therefore, we sequentially select a surrogate as the proxy of the victim model at each iteration. Then, the marginal contribution can be calculated based on the prediction of the proxy, which is related to the decline of its prediction logits (Eq.(2)) or cross-entropy loss increment (Eq.(3))

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over the ground-truth label of the adversarial example:

$$m(\hat{x}_S^t) = P(\hat{x}^{t-1}, y) - P(\bar{x}_S^t, y), \quad (2)$$

$$m(\hat{x}_S^t) = CE(logits(\bar{x}_S^t), y) - CE(logits(\hat{x}^{t-1}), y), \quad (3)$$

where $P(x, y)$ means the prediction probability with respect to label y for the input instance of x on the proxy victim model, $\hat{x}^t$ is the generated adversarial example at t-th iteration, and $\bar{x}_S^{t+1}$ represents the adversarial example generated by $\hat{x}^t$ for Shapley value computation in our simulated attack for all models in the set S at $(t+1)$-th iteration. Additionally, $CE(\cdot)$ indicates the cross-entropy loss.

Obviously, the computation of ϕ_i is akin to calculating the expectation of a random variable. As the exact estimation of Shapley value involves $N!$ permutations of all surrogate models, it is so expensive cost if N is large enough. Fortunately, the adversarial examples are usually generated offline, so we do not think it is detrimental to the transferability promotion of adversarial examples brought by Shapley value. Moreover, we only explore the naive estimation to demonstrate our effectiveness, as existing ensembles [4, 6] commonly adopt a four-surrogate strategy, which does not involve too much cost, which is experimentally confirmed later. If necessary, efficient estimations [12, 13] can be applied to tackle this issue.

Shapley Ensemble Adversarial Attack. Now, the weight of each surrogate model in an ensemble can thus be dynamically adjusted to promote the fusion of various models with different properties. To prevent excessive focus on a model in an iteration, we perform a scaling operation to update the weights of various surrogate models more smoothly, denoted as: $w_i^t = (1-\mu) \cdot w_i^0 + \mu \cdot \phi_i^t$, where w_i^0 is the initial weight of surrogate models and set as $1/N$, μ scales the effect of individual surrogates. Then combining all aforementioned equations, we can easily derive our proposed SEAA based on the existing strategy of SVRE [4], such as MI-FGSM-SEAA displayed as Algorithm 1 in the appendix.

4. EXPERIMENTS

4.1. Implementation Details

Dataset. For fair comparison with the recent adversarial attacks [4, 6], we also conduct our experiments on ImageNet.

Surrogates. We take into account of four classical neural networks as our surrogates to generate adversarial examples, including Inception-v3 (Inc-v3) [14], Inception-v4 (Inc-v4), Resnet-101 (Res-101) [15], and Inception-Resnet-v2 (IncRes-v2) [16]. To comprehensively compare the transferability, we first evaluate the attack performance against five normally trained models. In addition, adversarial training, input transformation-based defenses, and robust vision transformers can be utilized to evaluate the effectiveness of our adversarial examples. Particularly, we picked out three adversarially trained models, *i.e.*, Inc-v3$_{ens3}$, Inc-v3$_{ens4}$, and IncRes-v2$_{ens}$ [5], five transformation-based defenses adopted in SVRE [4] including R&P, NIPS-r3, JPEG, FD, and RS, and another three robust vision transformers containing RVT-S* [17], Drvit [18] and Vit+Dat [19].

Baselines. We integrate the Shapley value into SVRE [4], and then combine the proposed SEAA with different adversarial attack baselines, such as I-FGSM [9], MI-FGSM [2], TIM [3], and TI-DIM [3], to generate adversarial examples.

Parameters Settings. We only introduce μ to smooth the coefficient of Shapley value, whose value varies with the attack. Specifically, 0.7 is set in most cases except 0.2 for all the experiments of TI-DIM. The bound of adversarial perturbation ϵ is set as $16/255$ like most works. For fair comparisons, the remaining parameters are consistent with SVRE [4].

Base	Attack	VGG19	Dense	MNV2	MNV3	EffNt	Avg
I-FGSM	Ens	75.6	67.3	77.2	66.6	59.3	69.2
	SVRE	86.4	84.4	87.7	78.7	76.9	82.8
	SEAA	**90.5**	**89.2**	**91.3**	**83.6**	**82.8**	**87.5**
MI-FGSM	Ens	90.0	89.2	91.1	88.6	83.1	88.4
	SVRE	95.6	95.6	96.2	93.9	93.5	95.0
	SEAA	**96.7**	**96.8**	**97.0**	**94.8**	**95.0**	**96.1**
TIM	Ens	89.5	91.3	91.2	86.9	86.9	89.2
	SVRE	91.6	94.2	93.2	90.3	92.4	92.3
	SEAA	**93.6**	**96.4**	**94.8**	**91.7**	**94.5**	**94.2**
TI-DIM	Ens	93.9	95.3	94.5	92.8	93.3	94.0
	SVRE	95.4	97.3	96.3	94.6	97.1	96.1
	SEAA	**95.6**	**97.6**	**96.7**	**95.6**	**98.0**	**96.7**

Table 1. The attack success rate (%) of adversarial examples against normally trained models under the black-box setting.

4.2. Experimental Results

Attack Normally Trained Models. We first evaluate the performance of the proposed SEAA method on normally trained models, including VGG19 with batch normalization (VGG19) [20], DenseNet (Dense) [21], MobileNetV2 (MNV2) [22], MobileNetV3-Large (MNV3) [23] and EfficientNet-B7 (EffNt) [24]. These pre-trained models are well-known and easily accessible from the open sources provided by the authors. Note that the attacked models are not covered in the aforementioned surrogates. This is totally different from previous works [4, 6, 2], where one of the surrogate models is set as the target for the black-box attack while the remaining are utilized to craft adversarial examples.

The results evaluated by attack success rate are reported in Table 1. Note that the performances of Ens and SVRE are re-implemented by us with their released codes. After a deep analysis of these results, we can make an important conclusion: The proposed SEAA method can improve the performance of transferability for the adversarial examples generated by our ensemble attack. Specifically, the maximum gains of the SEAA method over Ens and SVRE are **23.5%** and **5.9%**, which occurred at the attack EffNt on the base of I-FGSM. The average boost under the black-box setting over

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Base	Attack	Adversarial Training				Transformation-based Defense					
		Inc-v3$_{ens3}$	Inc-v3$_{ens4}$	IncRes-v2$_{ens}$	Avg	R&P	NIPS-r3	JPEG	FD	RS	Avg
I-FGSM	Ens [2]	27.1	24.5	15.7	22.4	15.2	18.9	41.8	37.1	24.9	27.6
	SVRE [4]	40.1	37.3	24.8	34.0	25.1	32.9	67.1	48.3	27.0	40.1
	AdaEA [6]	35.4	31.8	18.6	28.6	17.9	28.0	66.8	53.9	27.1	38.7
	SEAA	**43.0**	**39.2**	**25.9**	**36.0**	**26.3**	**36.6**	**74.0**	**54.5**	**27.7**	**43.8**
MI-FGSM	Ens [2]	50.5	49.3	32.3	44.0	33.0	44.6	75.9	62.8	36.6	59.4
	SVRE [4]	64.5	59.0	39.1	54.2	41.5	59.5	89.9	82.8	38.5	62.4
	AdaEA [6]	45.5	41.4	24.2	37.0	26.1	41.9	81.4	73.6	35.7	51.7
	SEAA	**69.0**	**62.8**	**42.0**	**57.9**	**45.1**	**63.5**	**92.1**	**85.7**	**39.2**	**65.1**
TIM	Ens [2]	73.5	68.1	59.7	67.1	60.5	67.2	82.6	74.8	47.7	66.6
	SVRE [4]	87.9	85.6	79.7	84.4	78.7	83.0	91.8	92.1	61.8	81.5
	AdaEA [6]	82.1	79.7	74.0	78.6	72.6	76.9	85.0	87.7	59.9	76.4
	SEAA	**90.4**	**88.2**	**83.1**	**87.2**	**83.3**	**86.2**	**94.1**	**94.0**	**63.7**	**84.3**
TI-DIM	Ens [2]	87.4	84.3	77.6	83.1	81.2	85.7	91.7	84.3	57.6	80.1
	SVRE [4]	95.3	93.7	90.1	93.0	91.7	93.4	96.1	96.2	73.6	90.2
	AdaEA [6]	84.8	83.6	78.1	82.2	76.9	81.6	88.6	90.5	71.2	81.8
	SEAA	**95.5**	**94.3**	**90.8**	**93.5**	**92.6**	**94.3**	**96.5**	**97.2**	**74.9**	**91.1**

Table 2. Performance comparisons against various defenses. The results of SVRE [4] and AdaEA [6] are re-produced by us.

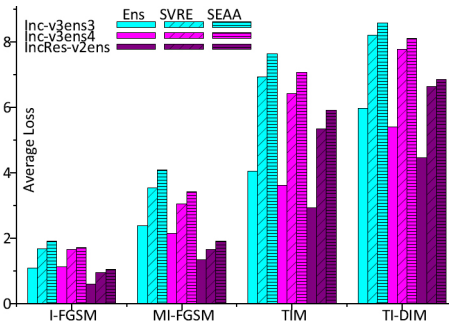


Fig. 1. The average loss of adversarially trained models with different ensembles.

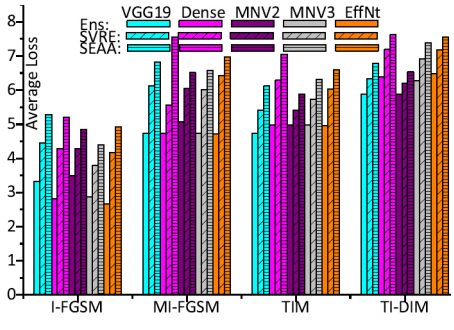


Fig. 2. The average loss of single black-box models via different ensembles.

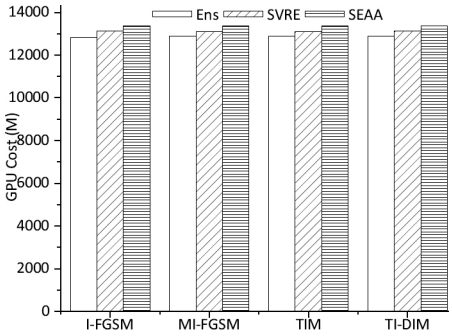


Fig. 3. GPU cost comparison of different ensemble strategies.

Base	Attack	Robust Vision Transformer			
		RVT-S*	Drvit	Vit+DAT	Avg
TIM	Ens [2]	52.8	69.1	40.7	54.2
	SVRE [4]	65.9	77.1	47.0	63.3
	AdaEA [6]	60.9	75.3	42.3	59.5
	SEAA	**73.4**	**78.3**	**55.1**	**68.9**
TI-DIM	Ens [2]	58.2	72.9	42.7	57.9
	SVRE [4]	75.7	84.5	54.7	71.6
	AdaEA [6]	63.4	77.8	44.7	62.0
	SEAA	**82.3**	**86.3**	**62.2**	**76.9**

Table 3. ASR comparisons against robust vision transformer.

SVRE ranges from **0.6%** to **4.7%**. We can attribute this significant improvement to the introduction of the Shapley value, which treats the ensemble as a cooperative game and properly evaluates the contribution of each surrogate to steer the generation of adversarial examples toward better transferability.

Adversarial Attack Defense Models. A variety of defense methods have been proposed to enhance the robustness of DNNs. Therefore, verifying the effectiveness of the SEAA method against those advanced defenses can better confirm our practicability and generalization. We report the performance against various advanced defenses in Table 2 and Table 3. By comparing these results comprehensively, some remarkable observations can be obtained:

(1) Adversarial training is extensively used as defense, while our SEAA approach can easily beat them, which can in turn promote the weakness identification of DNNs. Specifically, the biggest jump over SVRE [4] and AdaEA [6] can be more than **3%** and **20%** respectively, which occurs in the ensemble adversarial attack based on MI-FGSM.

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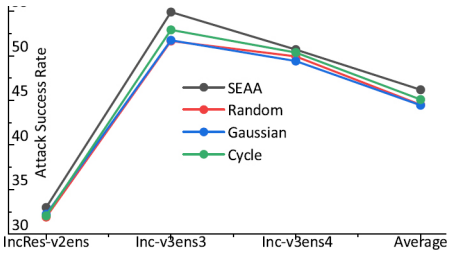


Fig. 4. Performance under different re-weights based on MI-FGSM.

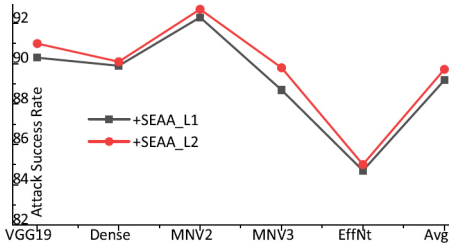


Fig. 5. Performance under different losses based on MI-FGSM.

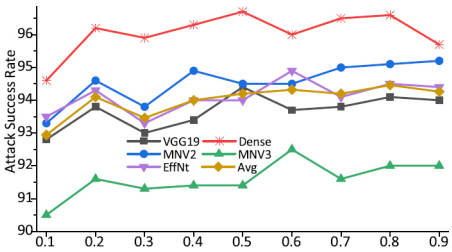


Fig. 6. Performance varying with different μ based on TIM.

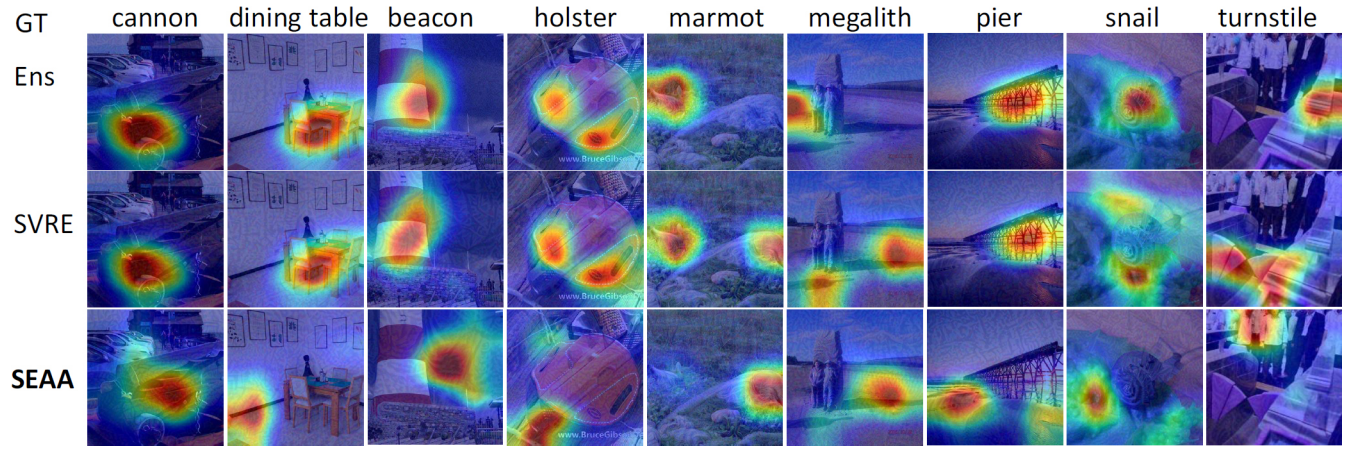


Fig. 7. Attention shift between different ensemble based on TIM [3]. The first row is the ground-truth label of raw image.

(2) Additionally, in all comparisons against input transformation-based defenses, the proposed SEAA method can also precede existing ensemble strategies, even SVRE by a clear margin from **0.4%** to **4.6%**. More importantly, our strategy can improve the ensemble adversarial attack based on the baseline TI-DIM to achieve an average success rate of over **90%** and a maximum of **97.2%** for black-box attacks even under the widely used defense mechanisms. This raises concerns about the deployment in security fields.

(3) Even the popular transformer with robust technology [17, 19, 18], we can still maintain a high attack success rate with the adversarial examples generated by an ensemble of only several classic surrogates. Under TI-DIM, the average attack success rate can reach **76.9%**, and we can get an optimum performance of **86.3%** when attacking Drvit [18]. Again, we ascribe this success to the introduction of the Shapley value into ensemble adversarial attack for better transferability. Additionally, this case shows that even the powerful transformers are compromised against adversarial examples, which emphasizes the robustness of DNNs.

4.3. Further Analysis

Loss Comparison. SVRE argued that the average loss over the generated adversarial examples can provide intuitive evidence of their transferability boost, that is, higher losses mean more intensified adversaries and better transferability on black-box attack [4]. Therefore, we indirectly confirm our effectiveness by comparing the average losses of different ensemble strategies. The models we concern with are the black-box setting of adversarially trained models and single models.

Figure 1 and Figure 2 show that our SEAA can obtain the biggest average loss on the aforementioned five baselines of attacks to synthesize well-transferred adversarial examples.

Ablation Study. To further demonstrate the superiority of our SEAA method, three re-weight strategies are considered to evaluate the contributions of different models: 1) Random means that the weights of all surrogates are randomly initialized at each iteration; 2) Gaussian indicates that the weights in each iteration follow a Gaussian distribution with an average value of $1/N$; 3) Cycle means a surrogate is selected to randomly increase its weight at each iteration. Obviously, the strategy of Cycle seems to be close to our SEAA method. To avoid randomness, the results reported in Figure 4 were also averaged over five times. Clearly, figure 4 shows that the proposed SEAA method occupies the top of the performance chain, and confirms the effectiveness of Shapley value in improving the transferability for ensemble attacks.

Additionally, we also explore two ways to compute the marginal value. L1 indicates that the prediction logits de-

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cline (Eq.(2)) is utilized to estimate the contribution of each surrogate, while L2 means that the cross entropy increment (Eq.(3)) is superimposed on top of L1. Figure 5 indicates that the latter strategy can help the Shapley value to better estimate the contribution of each sub-model, which promotes the generation of adversarial examples with better transferability.

Complexity Analysis. We also investigate the overhead of different ensemble strategies. When compared to SVRE, the extra requirement of GPU for our SEAA is less than $500M$, displayed in Figure 3. We consider this overhead acceptable as it does not cause much difficulty in the generation of adversarial examples. Furthermore, we are mainly concerned with the introduction of cooperative games to evaluate the contribution of each surrogate, thereby reweighing them at each iteration to synthesize better adversaries. Therefore, only naive Shapley value estimation is employed to verify this novel idea, while the aforementioned methods of Shapley value computation [13, 12] can also be utilized to reduce such an acceptable overhead.

Sensitivity of hyper-parameter μ. In this part, we conduct a series of experiments based on TIM to study the sensitivity of hyper-parameter μ. The setting is copied from the single black-box attack, displayed in Table 1. Figure 6 reveals that the optimal can not be uniformly obtained at a certain μ for all basis attacks. However, it is undeniable that the idea of cooperative games can improve the transferability of adversarial examples. The practitioners can adapt μ to better performance according to real applications, albeit we set it to 0.7 in all experiments yet without yielding the best performance.

Visualization. To better understand our effectiveness, we apply Grad-CAM to compare attention maps of adversarial examples generated by different ensembles on the basis attack of TIM. As shown in Figure 7, our SEAA method can more effectively make the model shift its attention to other irrelevant places, which accordingly misleads the target model.

5. CONCLUSION

In this work, we developed a novel ensemble strategy to better resolve the transferability of adversarial examples under the black-box setting. Specifically, we were the first to introduce cooperative games into the ensemble adversarial attack and then reweighed every surrogate model at each iteration by estimating its contribution based on Shapley value computation, which thus helped the ensemble to steer the generation of adversarial examples toward transferability enhancement. Finally, extensive experiments with achieving a new SOTA bar of ensemble adversarial attack, and solid ablation studies under different settings on the widely used benchmark demonstrated the effectiveness and superiority of the proposed Shapley ensemble adversarial attack by promoting the transferability based on various attack baselines. However, this work is heuristic and suffers from the extra requirement of GPU cost, and we will solve it in the future.

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